



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING – (24CI3201R)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	AI-Powered Self-Adaptive Cloud Application Monitoring and Intelligent Incident Response System	
S. No.	University ID	Name
1	2420030638	Rayapureddi Rishi SriCharan
2	2420030161	Yennam Sesank Reddy
3	2420090074	Ramannagari Harshath
Name of the Guide		Anugu Swapna

Abstract

Modern cloud-native applications operate in highly dynamic environments where system failures, performance degradation, and security threats can emerge without warning. Traditional monitoring tools rely on static thresholds and manual intervention, resulting in delayed incident detection, prolonged downtime, and significant revenue loss. As organizations increasingly migrate to microservices-based architectures deployed across multi-cloud environments, the volume and complexity of operational data have grown beyond the capacity of human operators to analyze effectively. This project addresses these critical challenges by presenting an AI-powered, self-adaptive monitoring platform that continuously observes cloud application health, detects anomalies in real time, and triggers intelligent automated responses to incidents before they escalate into major outages.

The proposed platform integrates machine learning models trained on historical system metrics, log data, and incident records to predict potential failures and identify root causes with high accuracy. Unlike conventional monitoring solutions that use fixed alert thresholds, this system employs adaptive algorithms that dynamically learn normal behavior patterns and adjust detection sensitivity based on changing workload conditions. Built following Agile development principles across structured sprints, the platform incorporates a fully automated DevOps pipeline with integrated security scanning, containerized microservices orchestrated through Kubernetes, and real-time observability dashboards powered by Prometheus and Grafana. The intelligent incident response engine automatically executes predefined remediation workflows such as auto-scaling resources, restarting failed services, isolating compromised containers, and escalating critical alerts to on-call engineers.

Beyond reactive monitoring, the platform emphasizes proactive resilience and continuous improvement through a complete MLOps lifecycle. The anomaly detection models are continuously retrained on new incident data, ensuring the system becomes more accurate and adaptive over time. Security incidents are handled with dedicated DevSecOps pipelines that automatically scan for vulnerabilities, enforce compliance policies, and generate audit trails for regulatory requirements. By combining artificial intelligence with modern software engineering practices, this platform aims to fundamentally transform how organizations monitor, manage, and protect their cloud infrastructure, reducing mean time to detection and mean time to resolution while minimizing the operational burden on engineering teams.

Signature of the Guide

HOD